

Smart Fan Speed Control System using Fuzzy Logic

MINI PROJECT REPORT

1. Team Details

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2. Detailed Problem Statement

Conventional fan systems operate manually with fixed speeds, leading to poor energy efficiency and a lack of adaptability. This project aims to design an intelligent system that:

- Automatically adjusts fan speed based on the number of people (crowd level).
- Uses distance measurement (ultrasonic sensor) to estimate crowd density.
- Provides smooth and efficient control using fuzzy logic.

3. Hardware Implementation

3.1 Components Used

- **Microcontroller:** Arduino Uno
- **Sensor:** Ultrasonic Sensor (HC-SR04)
- **Actuator:** DC Fan
- **Driver Circuit:** TIP122 Transistor, 1N4007 Diode, Resistors
- **Misc:** Breadboard, Jumper Wires, Power Supply

3.2 System Architecture

1. **Input:** Ultrasonic sensor measures distance to the nearest object/person.
2. **Processing:** Arduino maps distance to crowd levels and processes it using fuzzy rules.
3. **Output:** A PWM (Pulse Width Modulation) signal is sent to the fan motor.

3.3 Working Principle

The HC-SR04 sensor continuously measures distance.

- **Short Distance:** Indicates high crowd density.
- **Large Distance:** Indicates few or no people.

Behavioral Logic:

- No people detected → Fan OFF.
- Moderate crowd → Medium speed.
- High crowd → High speed.

3.4 Fuzzy Logic Design (Hardware)

Inputs:

- **Distance:** Far (Empty), Medium (Normal), Near (Crowded).

Outputs:

- **Fan Speed:** OFF, Low, Medium, High.

Rule Base:

1. IF Distance is **Far** THEN Fan **OFF**.
2. IF Distance is **Medium** THEN Fan **Medium**.
3. IF Distance is **Near** THEN Fan **High**.

3.5 Advantages & Limitations

- **Advantages:** Automatic operation, energy efficiency, low cost, and real-time response.
- **Limitations:** Ultrasonic approximation can be affected by obstacles, reflections, and sensor field-of-view limits.

4. Software Implementation (Simulation)

4.1 Technologies Used

- **Python:** Logic implementation.
- **Streamlit:** Web-based User Interface.
- **NumPy & Matplotlib:** Mathematical computations and graph plotting.

4.2 System Description

A web-based simulation developed to visualize how the fuzzy logic controller behaves under varying conditions.

- **Simulated Inputs:** Temperature (0-50°C) and Crowd (0-10 people).
- **Output:** Fan Speed (0-100%).

4.3 Fuzzy Logic Algorithm

1. **Fuzzification:** Convert crisp inputs into fuzzy linguistic values.
2. **Rule Evaluation (Mamdani):** Using AND (min) and OR (max) operators.
3. **Aggregation:** Combining all rule outputs into a single fuzzy set.
4. **Defuzzification:** Calculating the crisp output using the Centroid Method:

4.4 Membership Functions

[INSERT IMAGE: Membership Function Graphs for Temperature, Crowd, and Speed]

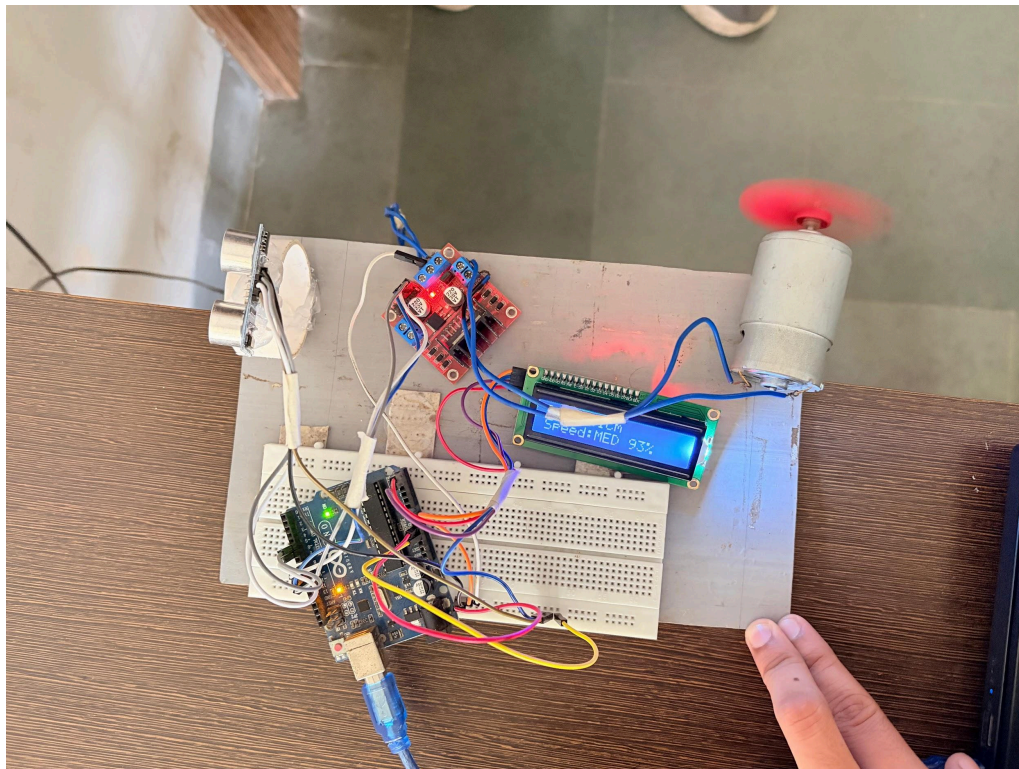
- Uses Triangular and Trapezoidal functions for smooth transitions.

5. Dataset Description

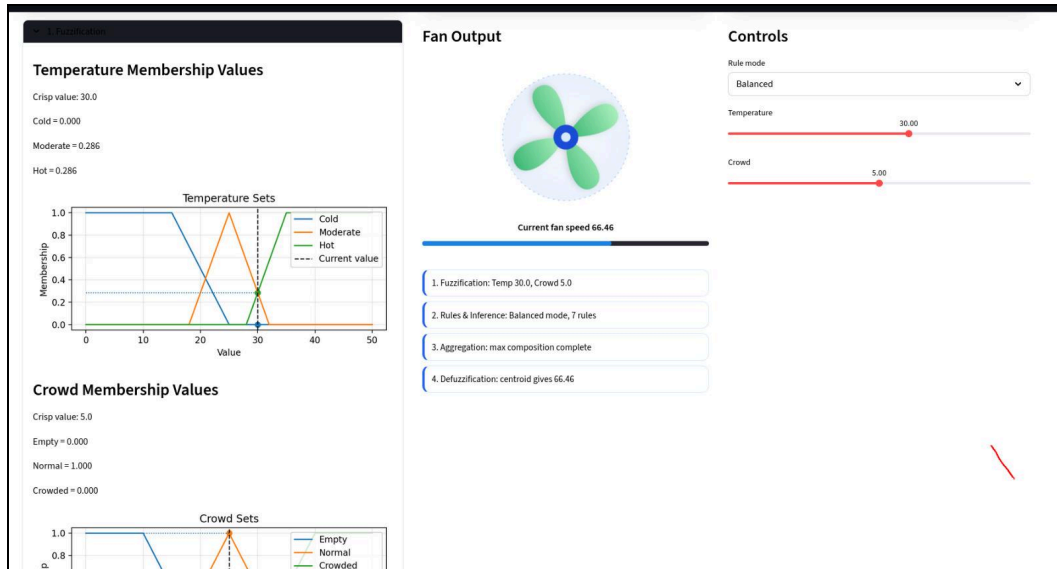
- **No static dataset used.** The system relies on real-time sensor data (Hardware) or user-provided sliders (Software Simulation).

6. Results / Screenshots

Hardware Performance



Software UI



7. Conclusion

The project successfully demonstrates an intelligent fan control system. Hardware provides real-time automation using ultrasonic sensing, while the software provides a visual understanding of the underlying fuzzy logic. The system effectively improves both user comfort and energy efficiency.

8. Links

- **GitHub Repository:** [link](#)

9. Future Scope

- **AI Integration:** Use camera-based person counting for higher accuracy.
- **IoT:** Integrate with a mobile app for remote monitoring.
- **Multi-Sensor Fusion:** Combine ultrasonic with PIR or Temperature sensors for better precision.